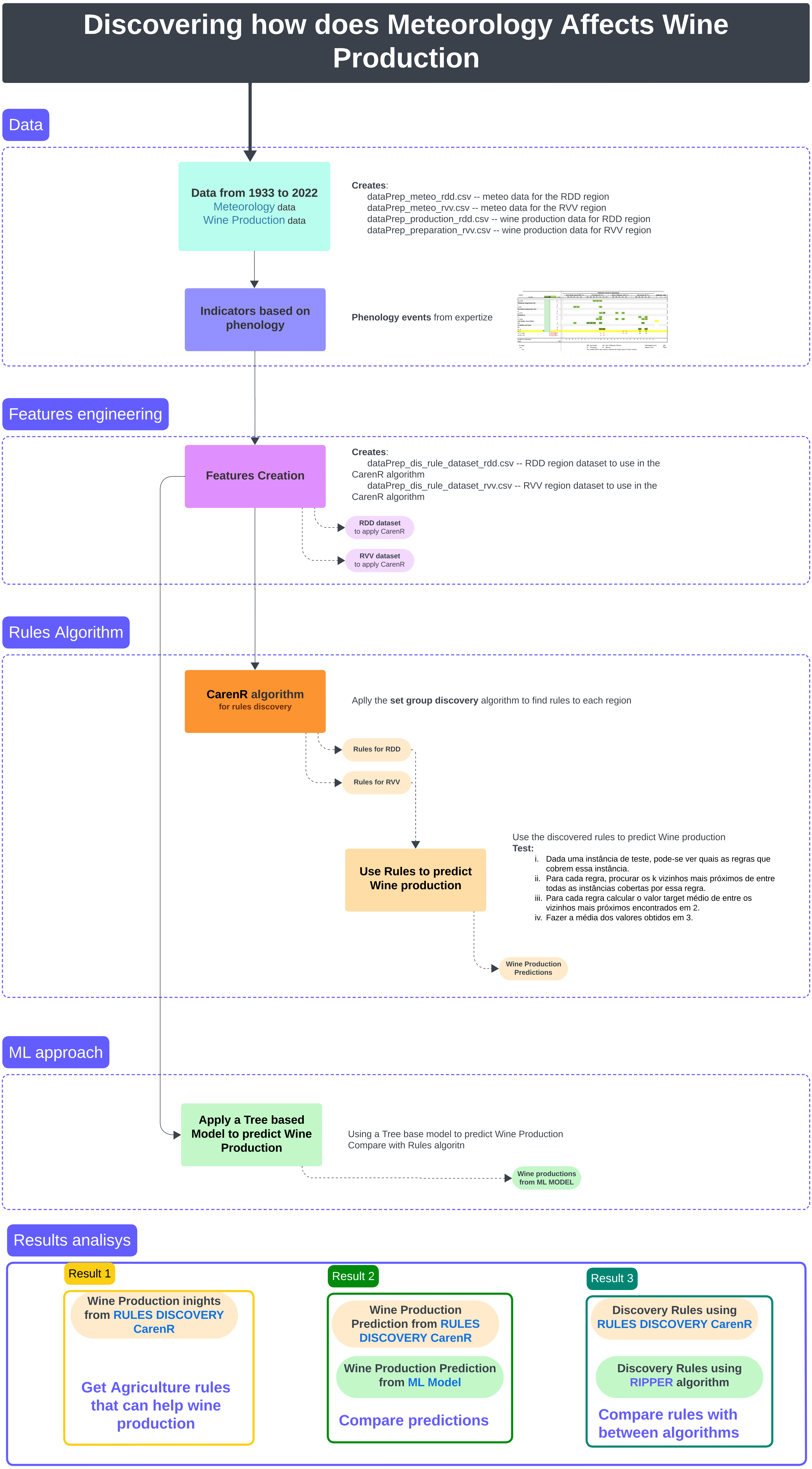
Discovering how does Meteorology Affects Wine Production
Data
Data from 1933 to 2022
Meteorology data
Wine Production data
Creates:
dataPrep_meteo_rdd.csv -- meteo data for the RDD region
dataPrep_meteo_rvv.csv -- meteo data for the RVV region
dataPrep_production_rdd.csv -- wine production data for RDD region
dataPrep_preparation_rvv.csv -- wine production data for RVV region
Indicators based on phenology
Phenology events from expertize
Features engineering
Features Creation
Creates:
dataPrep_dis_rule_dataset_rdd.csv -- RDD region dataset to use in the CarenR algorithm
dataPrep_dis_rule_dataset_rvv.csv -- RVV region dataset to use in the CarenR algorithm
RDD dataset to apply CarenR
RVV dataset to apply CarenR
Rules Algorithm
CarenR algorithm for rules discovery
Aplly the set group discovery algorithm to find rules to each region
Rules for RDD
Rules for RVV
Use Rules to predict Wine production
Use the discovered rules to predict Wine production
Test:
i. Dada uma instância de teste, pode-se ver quais as regras que cobrem essa instância.
ii. Para cada regra, procurar os k vizinhos mais próximos de entre todas as instâncias cobertas por essa regra.
iii. Para cada regra calcular o valor target médio de entre os vizinhos mais próximos encontrados em 2.
iv. Fazer a média dos valores obtidos em 3.
Wine Production Predictions
ML approach
Apply a Tree based Model to predict Wine Production
Using a Tree base model to predict Wine Production
Compare with Rules algoritn
Wine productions from ML MODEL
Results analisys
Result 1
Wine Production inights from RULES DISCOVERY CarenR
Get Agriculture rules that can help wine production
Result 2
Wine Production Prediction from RULES DISCOVERY CarenR
Wine Production Prediction from ML Model
Compare predictions
Result 3
Discovery Rules using RULES DISCOVERY CarenR
Discovery Rules using RIPPER algorithm
Compare rules with between algorithms